

[54/A-27]

**SARDAR PATEL UNIVERSITY****M.Sc. IT ( Integrated ) – IV<sup>th</sup> SEMESTER (CBCS)****PS04FIIT01 : Computer Based Numerical and Statistical Methods**

Date : 11/04/2017

Time : 10:00 AM to 1:00 PM

Max Marks : 70

Q:1 Write answers of following Multiple Choice Questions :

[10]

- [01] From the following \_\_\_\_\_ method is not iterative method.  
(A) Lagrange's (B) False position  
(C) Bisection (D) None of these
- [02] For real root of an equation  $x^3 - 2x - 5 = 0$ , the root lies between \_\_\_\_\_.  
(A) 0 and 1 (B) 1 and 2  
(C) 2 and 3 (D) None of these
- [03] From the following \_\_\_\_\_ method is the best method to obtain root of equation  $f(x) = 0$ .  
(A) False position (B) Newton's Raphson  
(C) Bisection (D) None of these
- [04] All the formulae of interpolation are based on the fundamental assumption that the given data can be expressed as a \_\_\_\_\_.  
(A) Equation (B) Algorithm  
(C) Polynomial (D) Extrapolation
- [05] The y depends on x can be written as \_\_\_\_\_.  
(A)  $f(x, y)$  (B)  $f(y, x)$   
(C)  $f(x)$  or  $y_x$  (D) All of these
- [06] The \_\_\_\_\_ is called the backward difference operator.  
(A)  $\emptyset$  (B)  $\cup$   
(C)  $\Delta$  (D)  $\nabla$
- [07] We can find solution of system of linear, algebraic equations using \_\_\_\_\_.  
(A) Gauss-Seidel method (B) Newton-Raphson method  
(C) Bisection method (D) None of these
- [08] The system of linear equation  $AX = B$  can be solved by matrix inversion method only if \_\_\_\_\_.  
(A)  $|A| = 0$  (B)  $|A| \neq 0$   
(C)  $A \neq 0$  (D) A is symmetric
- [09] Seasonal variation means the variation occurring within \_\_\_\_\_.  
(A) A number of years (B) Parts of a month  
(C) Parts of a year (D) Parts of a day
- [10] A time series is a set of data recorded \_\_\_\_\_.  
(A) Periodically (B) At time or space intervals  
(C) At successive points of time (D) All the above

Q:2 Answer the following short questions : Attempt Any Ten

[20]

- [01] What is Absolute Error?
- [02] Write stopping rules to obtain approximate solution for given non-linear equations.
- [03] Use the False Position method to obtain approximate solution of the equation  $x^3 - 9x + 1 = 0$ .
- [04] Write algorithm for Backward Difference Table.
- [05] What is Extrapolation?
- [06] Explain Central Difference Method.
- [07] List various Direct and Iterative Methods.
- [08] If  $x$  lies in the lower half of the table and if  $x_{k-1} < x < x_k$ , then what is  $\frac{dy(x)}{dx}$  and  $\frac{d^2y(x)}{dx^2}$ ?
- [09] If  $x$  lies in the upper half of the table and if  $x = x_k$ , then what is  $\frac{dy(x)}{dx}$  and  $\frac{d^2y(x)}{dx^2}$ ?
- [10] List Utilities of Time Series.
- [11] What do you mean by Cyclic Variation?
- [12] List Components of Time series.

Q:3 [A] Solve the following example using Bisection Method.

[06]

$X^3 - 4X - 9 = 0$  lies between 2.625 and 2.75. Find the root correct to three significant digits.

[B] Explain False Position Method.

[04]

OR

Q:3 [C] Solve the following example using Newton Raphson Method.

[06]

$X^3 - X - 4 = 0$  Find the root correct to three decimal places.

[D] Write algorithm for Iterative (Successive Approximation) Method.

[04]

Q:4 [A] Given the table of values are as follow :

[05]

|       |      |       |       |       |       |
|-------|------|-------|-------|-------|-------|
| X     | 2.0  | 2.25  | 2.50  | 2.75  | 3.0   |
| Y (X) | 9.00 | 10.06 | 11.25 | 12.56 | 14.00 |

Find Y (2.35) using Newton's Forward Formula.

[B] Given the table of values are as follow :

[05]

|       |   |   |   |    |
|-------|---|---|---|----|
| X     | 0 | 1 | 2 | 3  |
| Y (X) | 0 | 2 | 8 | 27 |

Find Y (2.5) using Lagranges Method.

OR

Q:4 [C] Given the table of values are as follow : [05]

|       |      |       |       |       |       |
|-------|------|-------|-------|-------|-------|
| X     | 2.5  | 3.0   | 3.5   | 4.0   | 4.5   |
| Y (X) | 9.75 | 12.45 | 15.70 | 19.52 | 23.75 |

Find  $Y(4.25)$  using Newton's Backward Formula.

[D] The following table gives the census population of a town for the years 1891 to 1931. Estimate the population for the year 1895 by using an appropriate interpolation formula. [05]

|                                  |      |      |      |      |      |
|----------------------------------|------|------|------|------|------|
| Year : X                         | 1891 | 1901 | 1911 | 1921 | 1931 |
| Population : Y<br>(in thousands) | 46   | 66   | 81   | 93   | 101  |

Q:5 [A] Solve the following system of equations using Matrix Inversion Method. [05]

$$2X_1 - 2X_2 + 5X_3 = 13$$

$$2X_1 + 3X_2 + 4X_3 = 20$$

$$3X_1 - X_2 + 3X_3 = 10$$

[B] The Distance(s) covered by a car in a given Time (t) is given in the following table : [05]

|                |    |    |    |    |    |
|----------------|----|----|----|----|----|
| Time (Minutes) | 10 | 12 | 16 | 17 | 22 |
| Distance (Km)  | 12 | 15 | 20 | 22 | 32 |

Find the Speed of the car at  $t = 14$  minutes.

OR

Q:5 [C] Explain the Gauss-Seidel Method for solution of system of linear equations. [05]

[D] Given the following table [05]

|            |      |      |      |      |      |
|------------|------|------|------|------|------|
| $x$        | 0.50 | 0.75 | 1.00 | 1.25 | 1.50 |
| $y = f(x)$ | 0.13 | 0.42 | 1.00 | 1.95 | 2.35 |

Find  $f'(0.75)$ .

Q:6 [A] Obtain the Trend from the Time Series given below by Method of Moving Average of 3 years and 5 years. [10]

| Year | Y   | Year | Y   |
|------|-----|------|-----|
| 1961 | 500 | 1966 | 560 |
| 1962 | 540 | 1967 | 600 |
| 1963 | 550 | 1968 | 640 |
| 1964 | 530 | 1969 | 620 |
| 1965 | 520 | 1970 | 640 |

OR

Q:6 [B] List Various Forecasting Models. Explain any Three in detail.

[10]